Isotopes and Average Atomic Mass

Isotopes, abundance and weighted averages

Name _____ Class _____ Date _____

METHOD

Average = Σ (fraction × isotope mass) Fraction = percentage ÷ 100

Give average atomic masses to two decimal places.

A. Understanding

1. What does the atomic mass shown on the periodic table represent?

2. An element has two isotopes, of mass 20.0 u and 22.0 u. Its average atomic mass is 20.2 u. Which isotope is more abundant? Explain.

B. Calculating average atomic mass

3. Boron is 19.9% boron-10 (10.013 u) and 80.1% boron-11 (11.009 u). Calculate its average atomic mass.

4. Rubidium is 72.17% rubidium-85 (84.912 u) and 27.83% rubidium-87 (86.909 u). Calculate its average atomic mass.

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5. Silver is 51.84% silver-107 (106.905 u) and 48.16% silver-109 (108.905 u). Calculate its average atomic mass.

6. Silicon has three isotopes: silicon-28 (27.977 u, 92.23%), silicon-29 (28.976 u, 4.68%) and silicon-30 (29.974 u, 3.09%). Calculate its average atomic mass.

C. Challenge

7. Copper has two isotopes, copper-63 (62.930 u) and copper-65 (64.928 u). Its average atomic mass is 63.546 u. Calculate the percentage abundance of each isotope.

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